SETUP GUIDE · HOME ASSISTANT VERSION

Home Assistant Setup Instructions

Connect a Mitsubishi Electric Ecodan heat pump to Home Assistant over MQTT

This guide takes you from a new unit to a working connection with your heat pump and your MQTT broker. Installation and setup take seven steps (**Part A**), followed by a check of the LED (Step 8). **Part B** contains safety, conformance and compliance information for the product.

Quick reference

Setup Wi-Fi network	`Mitsubishi Bridge AP`
Setup page (while on the setup network)	http://192.168.4.1 (the page may open automatically)
Device page (once on your Wi-Fi)	http://heatpump.local Change settings, restart, view status and install updates
Support	philip.thomson@f1p.uk

Status LED

Blue	Setup mode. Appears a few seconds after power-up, and the “Mitsubishi Bridge AP” network is available (Step 5).
Green	Connected to your Wi-Fi and to MQTT.
Orange	Connected to your Wi-Fi, but not to Home Assistant.

Button

Press	Restart the unit
Press and hold for 10 seconds	Reset the unit to its default settings

The button is located around the main LED.

What you will need

- The address, username and password of your MQTT broker (Home Assistant)
- Your Wi-Fi network name and password
- Access to the heat pump’s circuit breaker
- The cable supplied with the connector
- A phone or computer with Wi-Fi, to reach the setup page

PART A

Install and connect

1 Isolate the heat pump and open the indoor unit

SAFETY INFORMATION

Safely isolate your heat pump at the circuit breaker **before** removing the covers of the indoor unit. Follow the heat pump manufacturer's instructions.

Remove the covers of the indoor unit. Gently unplug any existing device (for example a MELCloud Wi-Fi adapter) from the red **CN105** connector on the heat pump circuit board.

2 Optional: reconnect your existing adapter

If you wish, connect the device you unplugged to the **MEL** port of your new connector.

3 Connect the unit to the CN105 connector

Take the cable supplied with the connector and plug it into the red **CN105** connector on the heat pump circuit board.

WARNING

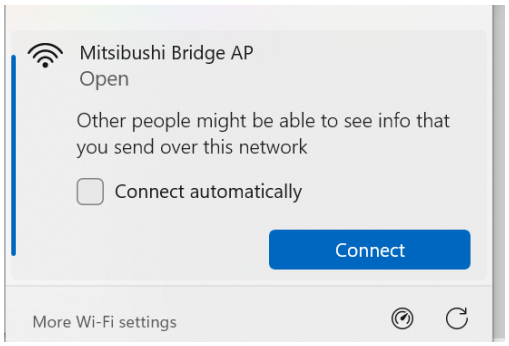
The **USB port (sealed off) must not be connected** at the same time as the heat pump.

4 Refit the covers and restore power

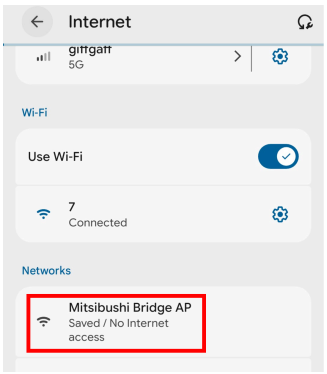
Fit the covers back on and restore power to your heat pump.

5 Connect to the setup network

After a few seconds the LED on the new unit shows **Blue** and the Wi-Fi network "**Mitsubishi Bridge AP**" becomes visible. Connect to it from your phone or computer.



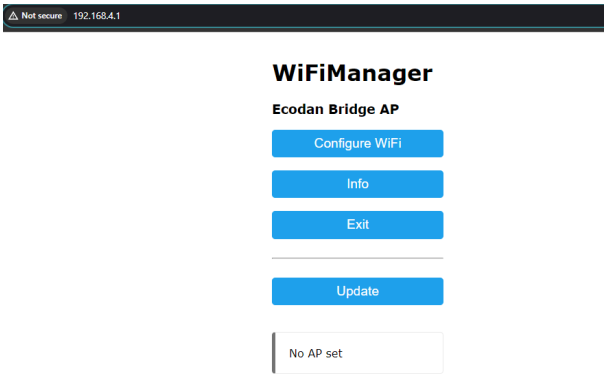
Example from Windows 11



Example from mobile

6 Open the setup page

Once connected, browse to http://192.168.4.1 and choose **Configure WiFi**. The page might open automatically.



7 Enter your details and save

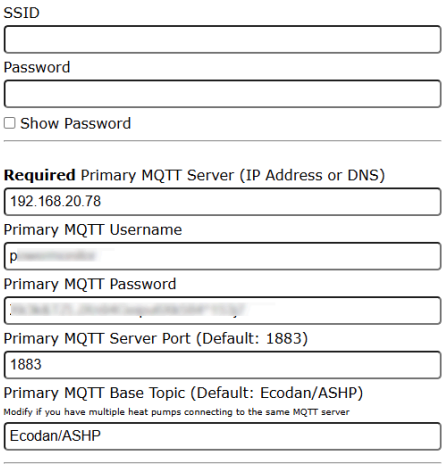
Select your Wi-Fi network and enter its password.

Enter your **Primary MQTT** details. The server address is required:

Server	IP address or DNS name
Username	Your MQTT username
Password	Your MQTT password
Port	Default `1883`
Base Topic	Default `Ecodan/ASHP`

Press **Save** at the very bottom of the page.

> **BASE TOPIC**
> Change the base topic only if you have several heat pumps connecting to the same MQTT server.



Setup form

8 Restart and check the LED

Restart the device after the first setup, then check the LED:

Green: the unit is connected to Wi-Fi and MQTT.

Orange: the unit is connected to Wi-Fi, but not to Home Assistant.

> **GOOD TO KNOW**
> **Local web page:** while the unit is on your Wi-Fi, visit http://heatpump.local to modify settings, restart the device, view status and perform updates.
>
> **Button:** press the button around the main LED to restart; press and hold for 10 seconds to reset to default.

PART B

Safety, conformance and compliance

Intended use and installation

The unit is a Wi-Fi and MQTT interface for Mitsubishi Electric Ecodan heat pumps. It connects to the red CN105 connector on the heat pump circuit board (Part A, Step 3). Installation requires removing the covers of the indoor unit: isolate the heat pump at the circuit breaker first and follow the heat pump manufacturer's instructions (Step 1). The sealed-off USB port must not be connected at the same time as the heat pump. If you are unsure at any point, stop and contact support.

Product and manufacturer

Product	Mitsubishi CN105 bridge Generation 2
Model / type reference	Proxy - Passive breakout board fitted with M5Stack AtomS3 Lite microcontroller
Firmware version	7.0.32
Manufacturer	Philip Thomson 7 Tippermuir Close, Perth, Scotland, PH1 1AD
Support	philip.thomson@f1p.uk

Radio information

Radio technology	Wi-Fi IEEE 802.11 b/g/n
Frequency band	2.4GHz
Other radios (for example Bluetooth)	Bluetooth - Not Used & Disabled in Firmware

Declarations of conformity

UK	The full text of the UK declaration of conformity is available at: https://github.com/F1p/Mitsubishi-CN105-Protocol-Decode/blob/master/documentation/Technical%20File/Declaration%20of%20Conformity.pdf
EU	The full text of the EU declaration of conformity is available at the following internet address: https://github.com/F1p/Mitsubishi-CN105-Protocol-Decode/blob/master/documentation/Technical%20File/Declaration%20of%20Conformity.pdf
Conformity marking	UKCA and/or CE

Applicable requirements and standards

Area	UK (Great Britain)	EU	Standards applied
Radio equipment: health and safety, EMC, spectrum use	Radio Equipment Regulations 2017 (SI 2017/1206)	Radio Equipment Directive 2014/53/EU	Refer to M5Stack Atom S3 Documentation
Cybersecurity	Product Security and Telecommunications Infrastructure Act 2022 and the 2023 Security Requirements Regulations	RED Article 3(3)(d), (e), (f): applies to products placed on the EU market from 1 August 2025	EN 18031-1:2024

Area	UK (Great Britain)	EU	Standards applied
Hazardous substances	RoHS Regulations 2012	Directive 2011/65/EU (RoHS)	EN IEC 63000
Waste electrical equipment	WEEE Regulations 2013	Directive 2012/19/EU (WEEE)	Product marked with the crossed-out wheeled bin symbol

Product security

Passwords	There is no password to access the first time setup network, once connected the local device webpage can be protected via Password set from the Home Assistant entity
Reporting a vulnerability	Report suspected security issues to philip.thomson@f1p.uk or via GitHub Issues. We aim to acknowledge reports within 10 working days and to keep you informed of progress.
Security updates	Security updates will be provided until at least January 2028. Firmware is installed from the unit's own web page (Update) at http://heatpump.local or via Home Assistant Notifications

Disposal and environment

Do not dispose of this product with household waste. Take it to a suitable collection point for waste electrical and electronic equipment (WEEE). This product complies with RoHS"

Open source and independence

This project is open source, and more information is available on GitHub:
https://github.com/F1p/Mitsubishi-CN105-Protocol-Decode

The product is independent. It is not made, approved or endorsed by Mitsubishi Electric Corporation or its group companies. Mitsubishi Electric, Ecodan and MELCloud are trademarks of their respective owners and are used here only to identify compatible equipment.

> **NEED HELP?**
> Support is available by email at philip.thomson@f1p.uk.